

Benchmarking LLMs with Pre-Extracted Features for MDS-UPDRS Finger Tapping Assessment

Gary Wang

Nordling Lab @ Dept. of Mechanical Engineering
National Cheng Kung University No. 1 University rd.,
Tainan 70101, Taiwan



Outline

- 1 Background
- 2 Methods
- 3 Results
- 4 Model Agreement Analysis
- 5 Discussion
- 6 Conclusion
- 7 Future Work
- 8 Thanks



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Background

- Parkinson's Disease (PD)
 - Neurological disorder affecting 11.8 million globally in 2021.
 - Characterized by bradykinesia (slow decremental movement).
- MDS-UPDRS Item 3.4: Finger Tapping (FT)
 - A rating scale (0–4) for diagnosing bradykinesia severity.
 - Assesses interruptions, slowing, and amplitude decrement.
- Previous Work
 - Single-model evaluation (ChatGPT-4) showed potential with few-shot prompting.
 - Best result: 61.1% accuracy, 0.39 MAE, inter-rater ICC(3,k) = 0.85.
- This Study
 - Benchmark **14 LLM configurations** across multiple model families.
 - Compare zero-shot vs. few-shot prompting strategies.
 - Evaluate using MAE, RMSE, Correlation, Weighted Kappa, and Wasserstein Distance.

Models Under Evaluation

- **Gemini 3 Flash** (Google)
 - Few-shot and zero-shot configurations
- **DeepSeek-R1** (DeepSeek)
 - 32B few-shot & zero-shot, 70B few-shot & zero-shot
- **Qwen 2.5** (Alibaba)
 - 7B few-shot & zero-shot, 32B few-shot & zero-shot, 72B few-shot & zero-shot
- **Qwen3** (Alibaba)
 - 30B few-shot & zero-shot

Each session scored **10 times** independently to capture model variability.





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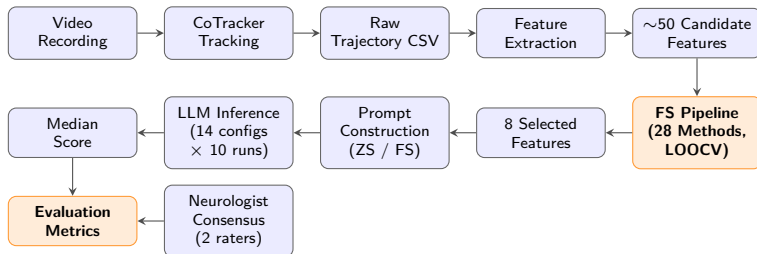
Prompt Design: This Study vs. Previous Work (Dallen 2024)

Prompt Element	Dallen 2024 (Prompt D)	This Study
Task description	✓	✓
MDS-UPDRS criteria	✓	✓
Raw trajectory CSV	✓ (10-tap, 5 cols)	×
Pre-extracted features	×	✓ (8 features)
Python analysis script	✓	×
Few-shot examples	✓ (TXT file, 4 scores)	✓ (embedded, 4 scores)
Approx. input tokens	~12,000–15,000	~633 (ZS) / ~1,525 (FS)
Token reduction	–	~8–23× fewer
Models evaluated	ChatGPT-4 only	14 LLM configurations
Best MAE	0.39	0.53
Best Kappa	–	0.54

ZS = zero-shot, FS = few-shot.



Experimental Pipeline



Features selected by FS pipeline (28 methods, LOOCV, $n=53$). See next slides for details.



Feature Selection Pipeline

Stage 1: Run **28 FS methods** (filter / embedded / wrapper) on 149 features

Stage 2: For each subset → **LOOCV** with fixed predictor (MLP-8, ReLU, Adam, $\alpha=1.0$)

Stage 3: Score with **4 metrics** (MAE, Spearman, Kappa, Adj. Acc.) + 2000-iter bootstrap CIs

Stage 4: Rank FS methods by **Composite** = mean of normalised metrics

Stage 5: Filter ($\text{Kappa} \geq 0.5$, $N_{\text{feat}} < 15$) → count **feature inclusion frequency** (17/28 pass)



Signal Preprocessing



Raw input: 2D coordinates of thumb & index finger per frame.

Distance	$\sqrt{(\Delta X)^2 + (\Delta Y)^2}$ (px)
Norm_Dist	$\frac{\text{Distance} - \min(d)}{\max(d) - \min(d)} \in [0, 1]$
Speed	$d(\text{Distance})/dt$ (px/s)
Speed_norm	$d(\text{Norm_Dist})/dt$
Accel.	$d(\text{Speed})/dt$ (px/s ²)
Accel._norm	$d(\text{Speed_norm})/dt$

Peak = frame of maximum finger separation (tap).

Valley = frame of minimum separation between taps.

Per-tap amplitude:

$$A_k = d(\text{peak}_k) - \frac{d(v_{\text{before}}) + d(v_{\text{after}})}{2}$$

Normalised amplitude:

$$\tilde{A}_k = \frac{A_k}{d_{\max} - d_{\min}}$$

removes hand-size & camera-distance confounds

Inter-tap interval (ITI):

$$\text{ITI}_k = t(\text{peak}_{k+1}) - t(\text{peak}_k)$$

Opening phase of tap k :

valley_{before} $\rightarrow$ peak _{k}

Closing phase of tap k : peak _{k} $\rightarrow$ valley_{after}





Feature Groups Overview — Unified Clinical Grouping (149)

Group	Clinical Domain	N
A. Kinematics	Raw finger displacement range & variability	48
B. Speed	Tapping rate, opening/closing velocity, and their decrement	35
C. Amplitude	Finger-opening size, shape, and progressive decrement	31
D. Rhythm	Temporal regularity of inter-tap intervals	14
E. Freezing	Movement pauses and halts	7
F. Task Completion	Ability to perform the full 10-tap sequence	5
G. Phase Asymmetry	Timing asymmetry between opening and closing	9
Total		149



Group A: Kinematics (48)



Feature	Definition
finger_mvmt_thumb_x_* (6)	$ \Delta X_{\text{thumb}} $, 6 stats
finger_mvmt_thumb_y_* (6)	$ \Delta Y_{\text{thumb}} $, 6 stats
finger_mvmt_thumb_dist_* (6)	$\sqrt{\Delta X_T^2 + \Delta Y_T^2}$, 6 stats
finger_mvmt_index_x_* (6)	$ \Delta X_{\text{index}} $, 6 stats
finger_mvmt_index_y_* (6)	$ \Delta Y_{\text{index}} $, 6 stats
finger_mvmt_index_dist_* (6)	$\sqrt{\Delta X_I^2 + \Delta Y_I^2}$, 6 stats
acceleration_* (6)	$ d^2(\text{Distance})/dt^2 $ (px/s ²), 6 stats
acceleration_norm_* (6)	$ d^2(\text{Norm_Dist})/dt^2 $, 6 stats

Stats: median, IQR, mean, min, max, std

Motivation

PD patients exhibit reduced range of motion and increased spatial variability due to bradykinesia and rigidity.

Thumb and index finger are tracked separately: thumb acts as the pivot; index executes the tap. Both contribute independent kinematic information.

Normalised acceleration captures relative sharpness; raw acceleration (px/s²) preserves absolute scale for cross-subject comparison.

Group B: Speed (35)

Feature

Definition

speed_* (6)	$ d(\text{Distance})/dt $ (px/s), 6 stats
speed_norm_* (6)	$ d(\text{Norm_Dist})/dt $, 6 stats
frequency_* (6)	$f_k = 1/ITI_k$ (Hz), 6 stats
freq_lr_slope	Slope of linear fit to f_k
freq_lr_fitness_r2	R^2 of frequency-time fit
freq_fit_min_degree	Fixed = 1 (linear model)
V_open_* (3)	$\max(\dot{d})$ opening (px/s), 3 stats
V_open_norm_* (3)	$\max(\dot{d})/(d_{\max} - d_{\min})$, 3 stats
V_close_* (3)	$ \min(\dot{d}) $ closing (px/s), 3 stats
V_close_norm_* (3)	$ \min(\dot{d}) /(d_{\max} - d_{\min})$, 3 stats
R_V_open	$\text{med}(V[-3 :])/\text{med}(V[3 :])$, opening
R_V_close	Same ratio for closing

Motivation

Tapping speed directly reflects bradykinesia. Progressive deceleration ($\text{freq_lr_slope} < 0$) is a key MDS-UPDRS 3.4 sub-criterion.

Normalised speed removes amplitude confound; raw speed (px/s) preserves absolute scale. $R_V_open/close$ measure velocity decrement analogous to amplitude decrement.



Group C: Amplitude — Statistics (1/2, total 31)

$$A_k = d(\text{peak}_k) - \text{mean}(\text{adjacent valleys})$$

Feature	Definition
amplitude_* (6)	6 stats of $\{A_k\}$ (px)
amplitude_cv	σ/μ of $\{A_k\}$
amplitude_entropy	Entropy of A_k histogram
amplitude_MAD/skew/kurtosis	Spread, asymmetry, tail
amplitude_norm_* (6)	6 stats of $\tilde{A}_k = A_k/(d_{\max} - d_{\min})$

Motivation

Reduced amplitude is the most salient MDS-UPDRS 3.4 criterion.

amplitude_* preserves absolute pixel scale; useful for within-subject analysis.

amplitude_norm_* normalises by each subject's full hand-opening range ($d_{\max} - d_{\min}$), removing camera-distance and hand-size confounds for cross-subject comparison.



Group C: Amplitude — Decrement (2/2, total 31)

Feature

Definition

amp_decr_slope

Slope of linear fit to A_k

amp_decr_fitness_r2

R^2 of the fit

amp_decr_end_to_mean

$A_{\text{last}}/\bar{A}$

amp_decr_last_to_first_half

$\bar{A}_{2\text{nd}}/\bar{A}_{1\text{st}}$

$A_{\text{early}}, A_{\text{late}}$

Median of first/last 3 taps

$R_{A_3_3}$

$A_{\text{late}}/A_{\text{early}}$

D_{A_pct}, D_{A_max}

% decrement, range

k_{collapse}

First k : $A_k < 0.5 A_{\text{early}}$

$\tilde{A}_{\text{median}}$

$\text{median}(A_k/A_{\text{early}})$

$\beta_{\tilde{A}}$

Slope of linear fit to

A_k/A_{early}

$CV_{\tilde{A}}$

σ/μ of A_k/A_{early}

Motivation

Progressive amplitude reduction (hypometria) is the core decrement criterion in MDS-UPDRS 3.4.

A_{early} as patient-specific baseline makes $R_{A_3_3}$, $\tilde{A}$, and k_{collapse} independent of absolute amplitude, enabling cross-subject comparison of *relative* decline.

Group D: Rhythm (14)

Feature	Definition
period_* (6)	$ITI_k = t(\text{peak}_{k+1}) - t(\text{peak}_k)$ in s, 6 stats
aperiodicity	Shannon entropy of FFT power
periodEntropy	Entropy of ITI histogram
periodVarianceNorm	$\text{var}(ITI) / \max(ITI)$
I_ITI	$\text{mean}(ITI_{k+1} - ITI_k)$
I_ITI_norm	$I_{ITI} / \bar{ITI}$
ITI_MAD	Median absolute deviation
ITI_skew	Skewness of ITI distribution
ITI_kurtosis	Kurtosis of ITI distribution

Motivation

aperiodicity and periodEntropy capture global irregularity from frequency vs. time domain.

I_{ITI} (analogous to MSSD) measures *short-term* instability between adjacent taps — more sensitive to sudden timing errors than std.



Groups E & F: Freezing (7) & Task Completion (5)

E — Freezing / Interruption

Feature	Definition
numInterruptions	Count of $ITI > 2 \times \text{median ITI}$
numFreeze	Count of $ITI > 3 \times \text{median ITI}$
maxFreezeDuration	Longest such ITI (s)
num_inter_norm	$\frac{\text{numInterruptions}}{\text{num_peaks}}$
num_freeze_norm	$\frac{\text{numFreeze}}{\text{num_peaks}}$
S_halt	$\sum \text{duration of interruption-length ITIs (s)}$
F_halt	$S_{\text{halt}} / \text{recording duration}$

Motivation

Interruptions defined by $ITI > 2 \times \text{median ITI}$ (MDS-UPDRS clinical criterion); $S_{\text{halt}}/F_{\text{halt}}$ capture total time lost to abnormally long inter-tap gaps.

MDS-UPDRS 3.4 asks for 10 taps. $F_{\text{incomplete}}$ provides a continuous version of the binary completion criterion.

F — Task Completion

Feature	Definition
num_peaks / N_{valid}	Total valid taps detected
C_10	1 if $N_{\text{valid}} \geq 10$, else 0
T_10	Time (s) from tap 1 to tap 10
F_incomplete	$\max(0, 1 - N_{\text{valid}}/10)$



Group G: Phase Asymmetry (9)



Each tap split into Opening ($v_{\text{before}} \rightarrow \text{peak}$) and Closing ($\text{peak} \rightarrow v_{\text{after}}$).

Feature	Definition
$T_{\text{open_*}}$ (3)	Opening phase duration (s), 3 stats
$T_{\text{close_*}}$ (3)	Closing phase duration (s), 3 stats
$R_{\text{OC_*}}$ (3)	$T_{\text{open}} / T_{\text{close}}$ per tap, 3 stats

Note: opening/closing velocity ($V_{\text{open/close}}$) is in Group B.

Motivation

PD patients often show asymmetric impairment between opening and closing.

$R_{\text{OC}} < 1$ indicates closing takes disproportionately longer — associated with difficulty in muscle relaxation.



Data Structure: Regressor Matrix and Regressand

Organise all $n=53$ sessions into regressor matrix $\mathbf{X}$ and regressand $\mathbf{y}$:

$$\mathbf{X} \in \mathbb{R}^{n \times p}, \quad \mathbf{y} \in \{0, 1, 2, 3, 4\}^n$$

Row $\mathbf{x}_i^\top$: $p=149$ kinematic features of session i ; y_i : consensus MDS-UPDRS score.

Session	mvmnt_x_mean	periodEntropy	period_iqr	num_peaks	...	y_i
S01	0.312	0.11	0.07	18	...	0
S02	0.178	0.43	0.24	13	...	2
S03	0.094	0.71	0.58	9	...	3
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$

Goal: Find a sparse index set $\mathcal{S} \subseteq \{1, \dots, p\}$ such that $\mathbf{X}_{\mathcal{S}}$ (selected columns) suffices for accurate prediction of $\mathbf{y}$.



28 Feature Selection Methods



Filter (12)

- Pearson
- Spearman
- Kendall
- Mutual Info
- F-regression
- mRMR
- Distance Correlation
- HSIC
- Fisher Score
- ReliefF
- FCBF
- Variance Threshold

Embedded (11)

- LASSO (Standard)
- ElasticNet
- Adaptive LASSO
- LARS
- RF Importance
- Extra Trees
- Gradient Boosting
- XGBoost
- Permutation Importance
- Stability Selection
- Boruta

Wrapper (5)

- RFECV (RF)
- RFECV (SVR)
- RFECV (Lasso)
- SFS Forward
- SBS Backward

Evaluation dataset: 53 finger-tapping sessions (16 PD subjects); $\mathbf{X} \in \mathbb{R}^{53 \times 149}$; $\mathbf{y}$ = consensus neurologist score (0–4).

FS Method Families: Objective Functions

All families operate on $(\mathbf{X}, \mathbf{y})$ but optimise different objectives. Output in all cases: a selection mask $\mathcal{S} \subseteq \{1, \dots, 149\}$.

Embedded — LASSO minimises:

$$\mathcal{L}(\beta) = \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \|\beta\|_1$$

Non-zero $\hat{\beta}_j \Rightarrow$ feature j selected.

ElasticNet adds $\frac{\alpha}{2} \|\beta\|_2^2$.

Tree methods (RF, XGBoost, ...) rank features by impurity reduction across all splits.

Wrapper — iterate over subsets, minimise CV error of a fitted estimator:

$$\mathcal{S}^* = \arg \min_{\mathcal{S}} \text{CV-MAE}(\mathbf{X}_{\mathcal{S}})$$

RFECV eliminates the feature with the smallest coefficient iteratively.

By-product: internal predictions $\hat{\mathbf{y}}_{\text{wrap}}$.



FS Method Evaluation Protocol

Fixed predictor (isolates FS effect from model choice): MLP(hidden=(8,), ReLU, Adam, $\alpha=1.0$), LOOCV $n=53$

For each $\mathcal{S}$: train MLP on $\mathbf{X}_{\mathcal{S}} \rightarrow$ LOOCV $\hat{\mathbf{y}} \rightarrow$ evaluate with:

$$\kappa_w = 1 - \frac{\sum_{i,j} w_{ij} p_{ij}}{\sum_{i,j} w_{ij} p_{i+} p_{+j}}, \quad w_{ij} = (i - j)^2$$

Stage 5 Filter — keep methods satisfying both:

- $\kappa_w \geq 0.5$
- $|\mathcal{S}| < 15$

$\Rightarrow$ **17 / 28** methods pass

Feature Inclusion Frequency:

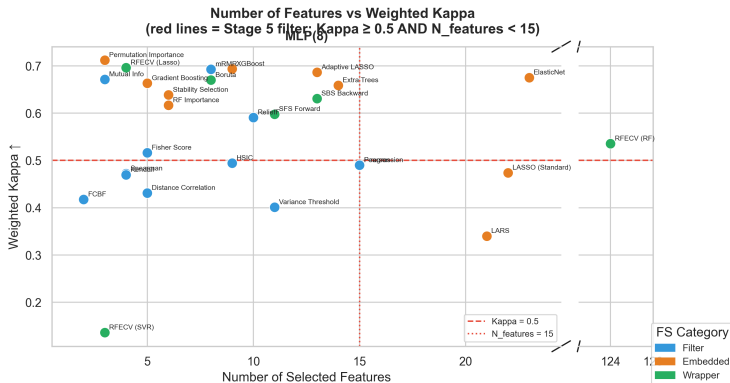
Count how many of the 17 passing methods select each feature.

Features chosen by $\geq 50\%$ form the final shortlist for LLM prompting.





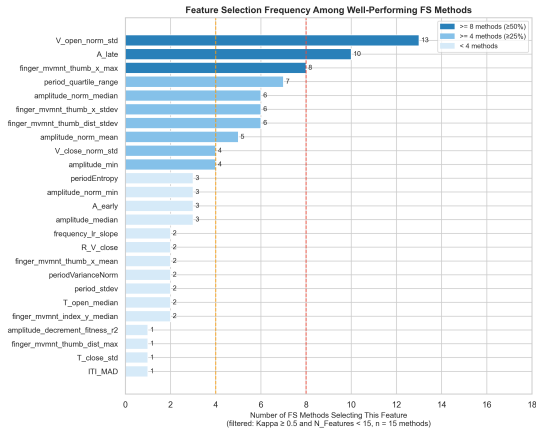
FS Pipeline: Compactness vs. Performance (149 features, $n=53$)



Red lines = Stage 5 filter (Kappa ≥ 0.5 , $N_{\text{feat}} < 15$); 17/28 methods pass.



FS Pipeline: Feature Inclusion Frequency



Only FS methods passing Stage 5 (17/28) contribute. Features selected by $\geq 50\%$ form the final shortlist.



Wrapper vs. MLP Prediction: Internal Score vs. Feature Quality

Wrapper methods produce internal CV predictions $\hat{y}_{\text{wrap}}$ using their own estimator; the standardised MLP on the same $\mathcal{S}$ gives $\hat{y}_{\text{MLP}}$.

Compare residuals (sorted by y_i):

$$\hat{y}_{\text{wrap},i} - \bar{y} \quad \text{vs} \quad \hat{y}_{\text{MLP},i} - \bar{y}$$

What a large gap means:

- Wrap bad, MLP **good**: features are genuinely informative but wrapper's estimator misjudges them
→ selection driven by noise risk
- Wrap good, MLP **bad**: wrapper overfits to its own estimator
- Small gap: internal score faithfully reflects subset quality

Motivates using a **fixed MLP** as the uniform yardstick.

Figure: per-session residuals

$$\hat{y}_{\text{wrap}} - \bar{y} \text{ (blue)}$$

$$\text{vs } \hat{y}_{\text{MLP}} - \bar{y} \text{ (orange)}$$

for each of the 5 wrapper methods.

(to be generated from FS pipeline logs)





Experimental Setup

- **Dataset:** 53 finger tapping sessions from 16 PD subjects.
- **Ground Truth:** Consensus score = average of two neurologists' ratings.
- **Input:** First 10 taps trajectory data extracted via CoTracker.
- **Prompt Strategies:**
 - **ZS:** MDS-UPDRS criteria + trajectory data only.
 - **FS:** Additional example feature sets from concordant sessions.
- **Evaluation Protocol:**
 - 10 independent evaluations per session per model.
 - Final score = median of 10 evaluations.





Evaluation Metrics

- **MAE** (Mean Absolute Error): Average magnitude of scoring errors. ↓
- **RMSE** (Root Mean Squared Error): Penalizes large deviations. ↓
- **Pearson Correlation**: Linear association with ground truth. ↑
- **Weighted Kappa**: Ordinal agreement penalizing distant disagreements. ↑
- **Wasserstein Distance**: Distribution-level scoring agreement. ↓
- **Stability**: Average SD across 10 evaluations per session. ↓
- **Accuracy (ACC)**: Fraction of exactly correct predictions (after rounding). ↑
- **Macro Recall**: Per-class recall averaged across all score classes (0–4). ↑

Additionally, **binary classification** (Normal=0 vs. Abnormal ≥ 1): ACC, Recall, Precision, F1.

All metrics reported with **95% bootstrap CI** (2,000 replicates).





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Performance Summary Table

Model	ACC ↑	AAcc ↑	MAE ↓	RMSE ↓	Corr ↑	Kappa ↑	WD ↓	Stab ↓
deepseek-r1_32b FS	56.6%	90.6%	0.53	0.85	0.56	0.54	0.26	0.449
deepseek-r1_32b ZS	37.7%	81.1%	0.81	1.09	0.37	0.30	0.58	0.658
deepseek-r1_70b FS	43.4%	90.6%	0.66	0.92	0.48	0.48	0.25	0.421
deepseek-r1_70b ZS	41.5%	98.1%	0.60	0.80	0.48	0.43	0.38	0.421
qwen3_30b FS	49.1%	88.7%	0.62	0.92	0.46	0.45	0.17	0.786
qwen3_30b ZS	45.3%	81.1%	0.74	1.06	0.30	0.30	0.06	0.876
qwen2.5_7b FS	32.1%	100.0%	0.68	0.82	0.52	0.45	0.45	0.620
qwen2.5_7b ZS	35.8%	98.1%	0.66	0.84	0.38	0.28	0.55	0.774
qwen2.5_32b FS	49.1%	88.7%	0.66	1.02	0.46	0.43	0.40	0.405
qwen2.5_32b ZS	49.1%	98.1%	0.53	0.75	0.55	0.50	0.34	0.588
qwen2.5_72b FS	50.9%	88.7%	0.64	1.01	0.48	0.44	0.42	0.314
qwen2.5_72b ZS	43.4%	98.1%	0.58	0.79	0.50	0.45	0.32	0.423
Gemini3 Flash FS	39.6%	86.8%	0.77	1.09	0.42	0.39	0.32	0.148
Gemini3 Flash ZS	49.1%	86.8%	0.66	1.00	0.49	0.37	0.58	0.331
Doctor (Tan)	60.4%	100.0%	0.40	0.63	0.72	0.69	0.25	0.000
Doctor (Chien)	64.2%	100.0%	0.36	0.60	0.85	0.83	0.32	0.000
Random (uniform)	3.8%	32.8%	1.46	1.82	~0.00	~0.00	0.74	–
Random (proportional)	17.2%	55.7%	0.96	1.27	~0.00	~0.00	0.13	–

AAcc = Acceptable Accuracy ($|\text{error}| \leq 1$).

Yellow = Doctor baselines (each doctor vs. consensus). Gray = Random baselines (10k trials).



Binary Classification: Normal (0) vs Abnormal (≥ 1)

Type	Model	ACC $\uparrow$	Recall $\uparrow$	Precision $\uparrow$	F1 $\uparrow$
ML	RandomForest	79.2%	89.2%	82.5%	85.7%
ML	XGBoost	73.6%	89.2%	76.7%	82.5%
Dallen	Gemini3 Flash ZS	75.5%	73.2%	93.8%	82.2%
LLM	DS-R1-32b FS	86.8%	100.0%	85.4%	92.1%
LLM	DS-R1-32b ZS	77.4%	100.0%	77.4%	87.2%
Doctor	Tan vs Consensus	84.9%	100.0%	93.2%	91.1%
Doctor	Chien vs Consensus	90.6%	90.2%	92.5%	93.7%
Random	Uniform	66.4%	80.0%	78.6%	78.6%
Random	Proportional	64.9%	77.4%	77.2%	77.2%

TML: Leave-One-Out CV (LOO-CV).

LLMs surpass both TML and random baselines; DS-R1-32b FS approaches doctor-level binary detection.





Baseline Context: Why Doctor & Random Baselines

■ Upper bound — Doctor baseline:

Matter

- Each doctor's score vs. consensus (average of both doctors).
- Doctor (Chien): Kappa = 0.83, Corr = 0.85 — the *ceiling* for any automated system.
- Doctor (Tan): Kappa = 0.69, Corr = 0.72 — shows inter-rater variability.

■ Lower bound — Random baseline:

- Uniform random: Kappa ≈ 0.00 , MAE = 1.46 — any model *must* beat this.
- Proportional random: MAE = 0.96 — even matching label distribution yields low accuracy.

■ LLM positioning:

- Best LLM (DS-R1-32b FS): Kappa = 0.54, sits at **65%** of the gap between random and doctor.
- All 14 LLM configs clearly surpass random baselines across all metrics.
- Gap to doctor level ≈ 0.15 – 0.30 Kappa — room for improvement.

Recall Analysis: Binary Classification

- **Binary Recall** ($\text{Abnormal} \geq 1$) is clinically critical:
 - Missing an abnormal case (false negative) has greater clinical cost than a false positive.
 - High binary recall = fewer missed abnormal sessions.
- **Key findings:**
 - DS-R1-32b (FS & ZS) achieves **100% recall** — no abnormal case missed.
 - Dallen's approach has lower recall (73.2%) — misses 11 out of 41 abnormal sessions.
 - DS-R1-32b FS achieves best balance: highest ACC (86.8%) with perfect recall.





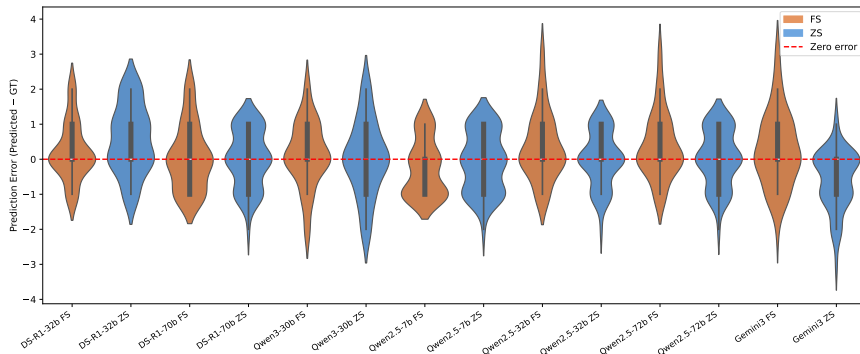
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Error Distribution per Model



Violin width = frequency of that error value. Red dashed line = zero error. Inner box = IQR & median.



Pairwise KS Test: Gemini3 Flash ZS vs All Models

Model	p -value
DS-R1-32b FS	4.2×10^{-4}
DS-R1-32b ZS	3.8×10^{-6}
DS-R1-70b FS	0.008
DS-R1-70b ZS	0.028
Gemini3 Flash FS	0.002
qwen2.5_7b FS	0.304
qwen2.5_7b ZS	0.049
qwen2.5_32b FS	2.8×10^{-5}
qwen2.5_32b ZS	0.028
qwen2.5_72b FS	7.3×10^{-5}
qwen2.5_72b ZS	0.028
qwen3_30b FS	0.004
qwen3_30b ZS	0.002

Two-sample KS test on prediction error distributions.

Red : $p < 0.05$ — Gemini3 Flash ZS is significantly different from 12 out of 13 models, indicating it is a distributional outlier.





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Model Size Effect: Qwen2.5 Scaling (ZS)

Model	ACC	AAcc	MAE	Kappa	Corr
qwen2.5_7b ZS	35.8%	98.1%	0.66	0.28	0.38
qwen2.5_32b ZS	49.1%	98.1%	0.53	0.50	0.55
qwen2.5_72b ZS	43.4%	98.1%	0.58	0.45	0.50

- 7b → 32b: clear improvement across all metrics.
- 32b → 72b: **performance drops** — larger is not always better.
- 32b appears to be the optimal size for this task under ZS.
- AAcc stays at 98.1% across all sizes — scaling affects *exact* accuracy, not tolerance.

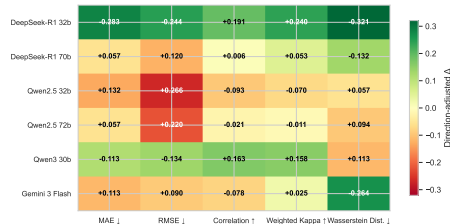


FS vs ZS: Effect Across All Models



Model	Δ Kappa	Δ MAE	Δ AAcc	Dir.
deepseek-r1_32b	+0.24	-0.28	-7.5%	FS
qwen3_30b	+0.15	-0.12	+7.6%	FS
qwen2.5_7b	+0.17	-0.02	+1.9%	FS
deepseek-r1_70b	+0.05	-0.06	-7.5%	FS
Gemini3 Flash	+0.02	-0.11	0%	mixed
qwen2.5_72b	-0.01	+0.06	-9.4%	ZS
qwen2.5_32b	-0.07	+0.13	-9.4%	ZS

Δ = FS - ZS. Green = FS better; Red = ZS better.



Green = few-shot better; Red = zero-shot better

Reasoning Distillation: Numerical Comparison

Model	Prompt	MAE ↓	Kappa ↑	Corr ↑	WD ↓	Stab ↓
Qwen2.5-32b (base)	FS	0.66 [0.47, 0.87]	0.43 [0.15, 0.64]	0.46 [0.18, 0.67]	0.40 [0.21, 0.64]	0.405
Qwen2.5-32b (base)	ZS	0.53 [0.38, 0.66]	0.50 [0.33, 0.64]	0.55 [0.37, 0.69]	0.34 [0.17, 0.51]	0.588
DS-R1-32b (distilled)	FS	0.53 [0.36, 0.72]	0.54 [0.29, 0.71]	0.56 [0.32, 0.73]	0.26 [0.17, 0.47]	0.449
DS-R1-32b (distilled)	ZS	0.81 [0.60, 1.02]	0.30 [0.10, 0.47]	0.37 [0.13, 0.56]	0.58 [0.36, 0.83]	0.658

Δ Kappa(distilled FS vs base FS): +0.11 Δ Kappa(distilled ZS vs base ZS): -0.20

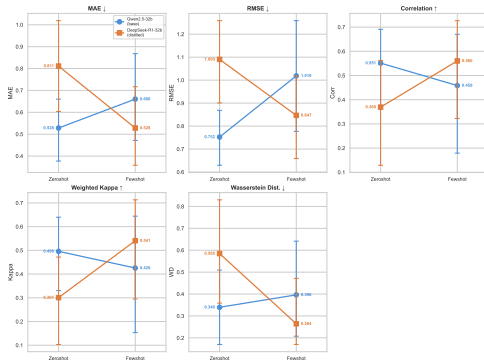
- Distilled + FS: best Kappa (0.54) and Corr (0.56) — few-shot examples anchor CoT reasoning.
- Distilled + ZS: worst MAE (0.81) and Kappa (0.30) — unconstrained CoT drifts without examples.





Effect of Reasoning Distillation

- DeepSeek-R1-32b = **Qwen2.5-32b** fine-tuned on DeepSeek-R1 CoT traces (RL reasoning distillation).
- **FS**: large gains (Kappa +0.11, MAE -0.13) — CoT anchored by examples.
- **ZS**: distillation *hurts* (Kappa -0.20, MAE +0.28) — unconstrained CoT drifts.
- **Implication**: Reasoning-distilled models need FS to realise potential.



Lines cross between Zeroshot and Fewshot, revealing the interaction effect.



Feature-Based vs Trajectory-Only Comparison

This work: feature extraction + reduced tokens. Dallen: raw trajectory (full tokens, ZS). Same 53 sessions.

	Model	ACC $\uparrow$	AAcc $\uparrow$	MAE $\downarrow$	Corr $\uparrow$	Kappa $\uparrow$	WD $\downarrow$	Stab $\downarrow$
Dallen	Gemini3 Flash (ZS)	45.3%	88.7%	0.70	0.44	0.39	0.49	0.41
This work	Gemini3 Flash ZS	49.1%	86.8%	0.66	0.49	0.37	0.58	0.331
This work	DS-R1-32b FS	56.6%	90.6%	0.53	0.56	0.54	0.26	0.449
This work	Qwen2.5-32b ZS	49.1%	98.1%	0.53	0.55	0.50	0.34	0.588
Baseline	Doctor (Chien)	64.2%	100.0%	0.36	0.85	0.83	0.32	0.000
Baseline	Random (uniform)	3.8%	32.8%	1.46	~ 0	~ 0	0.74	—

- **Same model (Gemini3 Flash):** feature extraction improves ACC 45.3% $\rightarrow$ 49.1%, Corr 0.44 $\rightarrow$ 0.49.
- DS-R1-32b FS achieves best overall: ACC 56.6%, Kappa 0.54 — far exceeding trajectory-only.
- Feature extraction reduces input tokens $\sim 28\times$ — lower cost, faster inference, compatible with smaller context-window models.





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Conclusion

- All models achieve MAE 0.53–0.81; **multiple LLM families** are capable of MDS-UPDRS FT scoring from trajectory data.
- **Best configurations:** DeepSeek-R1-32b FS (Kappa 0.54, Corr 0.56), Qwen2.5-32b ZS (MAE 0.53, RMSE 0.75).
- **Reasoning distillation is prompt-sensitive:** DeepSeek-R1-32b (= Qwen2.5-32b + CoT distillation) gains +0.11 Kappa with FS but loses −0.20 Kappa without examples — unconstrained chain-of-thought drifts.
- **Scaling benefit depends on base architecture and prompt strategy:** not a universal rule.
- **Stability varies widely:** 0.148 (Gemini) to 0.876 (Qwen3-30b ZS); consistency must be evaluated alongside accuracy.
- **No single model dominates:** choice should balance the target metric, available prompt budget, and consistency requirements.



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Future Work

- Evaluate additional LLMs (e.g., Claude, Llama 3, GPT-4o) for broader benchmarking.
- Investigate **structured CoT prompting** and **self-consistency** to mitigate reasoning drift observed in ZS distilled models.
- Expand dataset size and score distribution balance.
- Develop **ensemble methods** combining predictions from multiple LLMs.
- Explore **fine-tuning** smaller models on FT scoring data.





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Thanks for listening





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Appendix: Evaluation Metrics Definitions

- Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- Quadratic Weighted Kappa:

$$\kappa_w = 1 - \frac{\sum_{i,j} w_{ij} p_{ij}}{\sum_{i,j} w_{ij} p_{i+} p_{+j}}, \quad w_{ij} = (i - j)^2$$

- Macro Recall (averaged over K classes):

$$\text{Recall}_{\text{macro}} = \frac{1}{K} \sum_{k=1}^K \frac{\text{TP}_k}{\text{TP}_k + \text{FN}_k}$$

Appendix: Wasserstein Distance

Minimal work to transform one distribution into another.
Work = distribution weight moved $\times$ distance moved.
Lower WD = better agreement.

$$WD(P, Q) = \frac{\sum_{i=1}^M \sum_{j=1}^N d_{ij} f_{ij}^*}{\sum_{i=1}^M \sum_{j=1}^N f_{ij}^*}$$

Parameters:

- d_{ij} : distance between score categories i and j
- f_{ij}^* : optimal mass flow from category i to j
- M, N : number of score categories in distributions P and Q

